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Who Really Keeps Dancing? A Bayesian MCMC Inference and Logistic Dynamic Scoring Framework in Voting Systems

Summary

While Dancing with the Stars celebrates the artistry of the ballroom, a fundamental question persists: Who really keeps dancing?To quantitatively evaluate competitive fairness and its underlying drivers, we develop a comprehensive modeling framework and propose a **dynamic scoring system**.

First,we develop a **Bayesian MCMC inference model** under **hierarchical constraints** to address the unobservability of fan voting .The model incorporates a **Dirichlet prior**. Specifically, a **sparse prior** ($\alpha = 0.5$) is utilized to characterize regular seasons , while a **uniform prior** ($\alpha = 1$) is adopted in All-Star seasons. And three competition rules are transformed into truncation constraints within the likelihood function , with posterior sampling conducted via the **Metropolis-Hastings algorithm**. Results demonstrate that the model achieves **100% backtesting consistency** with historical outcomes ; furthermore, **posterior entropy** is employed to quantify the certainty of fan vote share inferences , with data indicating a steady upward trend in certainty throughout each season.

Then, comparing different scoring methods, we conclude that under the **percentage system**, fan preferences align with final results **25.7%** more consistently than under the ranking system.While the percentage system enhances fan engagement, its amplification of extreme popularity leads to anomalous advancements of low-skill contestants like Bobby Bones. Introducing a **judge rescue mechanism** significantly corrects this technical threshold.

Regarding the attribution of success and failure factors, we develop a **Lasso-based multi-factor influence attribution model**, employing **L1 regularization** for feature selection. The results reveal that judges' scores are significantly driven by the professional partners**(+3.97)** while exhibiting a bias against older contestants **(-2.19)**. In contrast, fan voting behavior demonstrates high inclusivity toward age **(-0.01)**. Furthermore, the study identifies a **"Social Media Paradox"**: while YouTube Views contributes slightly**(+0.007)** to voting, Social Media Personality is negatively**(-0.026)** correlated with actual votes. This reflects the presence of "slacktivism" among fan groups and a potential resistance toward emerging influencer culture among the television audience.

Finally we propose a **Logistic Dynamic Weighted System**. This system employs concave functions to apply nonlinear transformations to raw vote counts, thereby suppressing popularity monopolies, and incorporates a logistic weight curve that evolves over the course of the season. It effectively balances program entertainment value with professional fairness, providing a scientific decision-making pathway for competition mechanisms that integrate expert judging with collective intelligence.

Keywords: Bayesian Inference; MCMC; Lasso Regression; Counterfactual Analysis; Logistic Dynamic Weighted System

Contents

1 Introduction

1.1 Problem Background

Dancing with the Stars represents a complex Multi-Criteria Decision Making (MCDM) arena, where contestant outcomes are jointly determined by professional technical scores and subjective fan votes. However, the inherent tension between expert rationality and public sentiment frequently triggers "systemic controversy," leading the production team to evolve the scoring rules across 34 seasons—shifting from rank-based and percentage-based systems to the recent Judges Save mechanism.

Crucially, while judging scores are public, the actual "number of fan votes" remains confidential commercial information. This creates a significant inverse-inference challenge: reconstructing the hidden distribution of public preference from observed elimination sequences. By leveraging historical data from 34 seasons, this study develops a robust estimation framework to uncover the mathematical logic governing the interplay between fairness and entertainment value in this unique competitive ecosystem.

Figure 1: Show Poster

1.2 Restatement of the Problem

To decode the hidden dynamics of the DWTS scoring system, we structure our research into four integrated phases:

- **Phase I: Latent Variable Inference.** We develop an inverse probabilistic model to reconstruct unobservable fan vote shares, grounding our estimates in historical elimination outcomes and expert judge scores.
- **Phase II: Mechanism Comparative Study.** We evaluate the Rank-based versus Percentage-based systems through counter-factual simulations of controversial cases, assessing the fairness and inherent biases of each regime.
- **Phase III: Multi-dimensional Correlation.** We quantify the influence of extrinsic contestant attributes—such as industry background, age, and professional partners—on both technical merit and public popularity.
- **Phase IV: System Optimization.** We propose a refined hybrid scoring framework to harmonize technical rigor with entertainment value, supported by an actionable policy memo for show producers.

1.3 Our Work

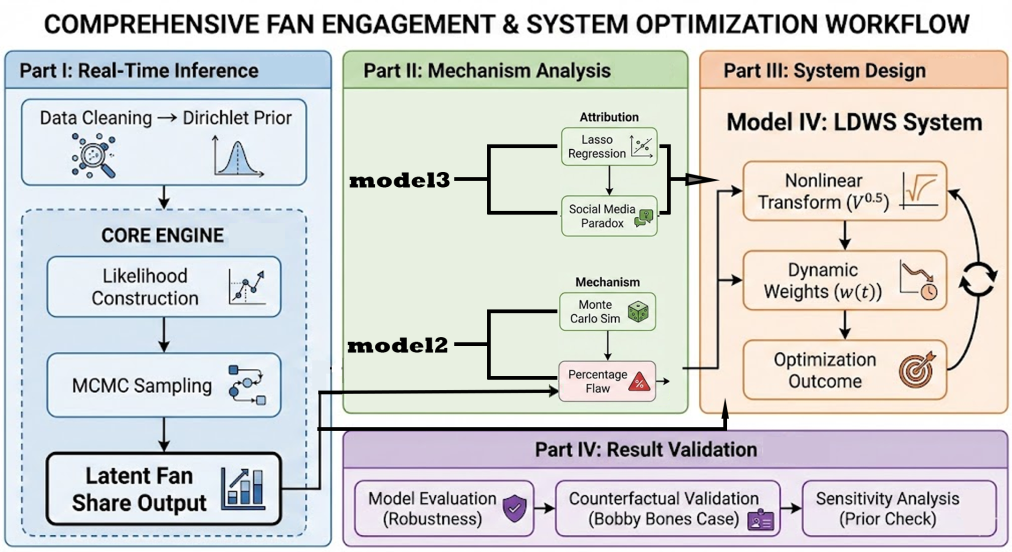

Figure 2: Overview of our framework

2 Assumptions and Explanations

To ensure mathematical tractability and logical consistency, we adopt the following assumptions:

Assumption 1: **Rule Consistency Post-Season 28.** We assume that starting from Season 28, the production team reinstated the integrated ranking method combining professional judges' scores and public fan votes.
Explanation: This assumption enables us to apply a consistent rule-based likelihood formulation for post-Season 28 data.

Assumption 2: **Independence assumption.** Weekly voting is treated as an independent statistical event.
Explanation: This assumption simplifies the likelihood construction by avoiding a dynamic dependence structure across weeks.

Assumption 3: **Rational elimination assumption.** Unless a *Judges' Save* event occurs, the eliminated contestant must be the one with the lowest combined score (or the worst combined rank) in that week.
Explanation: This links the observed elimination outcome to the latent vote-share vector and judges' information via rule-based constraints.

Assumption 4: **Zero-sum assumption.** Fan vote shares sum to a constant within each week, i.e., $\sum_{i\in\Omega_t} \pi_{i,t} = 1$.
Explanation: This reflects that vote shares are normalized proportions on a simplex, which motivates the Dirichlet prior.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Definition	Unit/Range
Sets & Indices		
t	Index of week	$\{1, 2, \ldots, T\}$
i	Index of contestant	$\{1, 2, \ldots, N\}$
Ω_t	Set of active contestants at week t	-
m	MCMC posterior sample index	$\{1, 2, \ldots, M\}$
M	Number of MCMC posterior samples	$\mathbb{Z}^+$
N_t	Contestants at week t	$\mathbb{Z}^+$
$E_{\text{obs},t}$	Observed eliminated set at week t	-
N_{judges}	Number of judges	$\{3, 4\}$
Judge Data		
$O_{i,t}$	Observed judge score	$[0, 30]$ or $[0, 40]$
$S_{i,t}$	Standardized judge score	$[0, 40]$
$J_{i,t}$	Judge score share	$[0, 1]$
$R_{i,t}^{\text{Judge}}$	Judge rank	$\mathbb{Z}^+$
Fan Data & Latent Variables		
$\boldsymbol{\pi}_t^{(m)}$	Fan vote distribution at week t	$\sum_i \pi_{i,t}^{(m)} = 1$
$\widehat{\boldsymbol{\pi}}_t$	Posterior mean fan vote share vector at week t	$\sum_i \widehat{\pi}_{i,t} = 1$
$H(\boldsymbol{\pi}_t)$	Posterior entropy of fan votes at week t	$\mathbb{R}^+$
$H_{\max}$	Maximum entropy of week t (i.e., $\ln(N_t)$)	$\mathbb{R}^+$
γ_t	Certainty score (normalized entropy deficit)	$[0, 1]$
$\pi_{i,t}$	Fan vote share	$[0, 1]$
α	Dirichlet concentration parameter	$\mathbb{R}^+$
$R_{i,t}^{\text{Fan}}$	Fan rank	$\mathbb{Z}^+$
Constraints & Status		
$I_{i,t}$	Withdrawal indicator	$\{0, 1\}$
$E_{i,t}$	Elimination indicator	$\{0, 1\}$
$E_{\text{pred},t}^{(m)}$	Simulated eliminated set (sample m) at week t	-
$C_{i,t}$	Combined score/rank	$\mathbb{R}$
$L(\cdot)$	Likelihood function	-

*There are some variables that are not listed here and will be discussed in detail in each section.

4 Data Collecting and Preprocessing

4.1 Data Collecting and Restructuring

We transform the raw competition records into panel data keyed by the composite key (i, t), where i denotes the i-th contestant and t denotes the competition week. On this basis, we introduce external data sources and use left joins to map contestants' static attributes (e.g., age, occupation, height, and partner identity) onto the dynamic competition data, yielding a dataset with $N = 2777$ observations.

Table 2: Data and Database Websites

Database Names	Database Websites
DWTS Celebrity Height Data	https://www.celebrityheights.com https://en.wikipedia.org
YouTube Viewer Data	https://socialblade.com https://vidiq.com
DWTS Contestant Data	https://en.wikipedia.org https://www.dancingwiththestars.com

4.2 Score Normalization

Because the number of judges N_{judges} may vary from 3 to 4 across seasons or weeks, the maximum possible total score (i.e., the scale) is not consistent. To remove this unit mismatch, we adopt a mean-imputation adjustment and convert all scores to the 4-judge standard.

Let $O_{i,t}$ denote the observed total judge score of contestant i in week t. The adjusted (standardized) total score $S_{i,t}$ is defined as

$$S_{i,t} = \begin{cases} O_{i,t} + \dfrac{O_{i,t}}{3}, & N_{\text{judges}} = 3, \\ O_{i,t}, & N_{\text{judges}} = 4. \end{cases} \tag{1}$$

Based on $S_{i,t}$, we further compute the relative judge-score share $J_{i,t}$ of contestant i in week t as a core performance indicator, where Ω_t denotes the set of all active contestants in week t:

$$J_{i,t} = \frac{S_{i,t}}{\sum_{k\in\Omega_t} S_{k,t}}. \tag{2}$$

4.3 Feature Engineering

To address the mixed-input setting of nominal and numerical variables and to eliminate the impact of unit scales on the Lasso penalty, we preprocess the explanatory variables as follows.

4.3.1 Categorical One-Hot Encoding

For discrete categorical variables such as *Industry*, *Partner*, and *Homestate*, we apply one-hot encoding to convert them into binary dummy variables. Let a categorical variable C have K levels. For an observation x, its encoding vector $\mathbf{v} \in \{0,1\}^K$ satisfies: if x belongs to category k, then $v_k = 1$; otherwise $v_k = 0$. This step expands the original feature space into a high-dimensional sparse matrix, enabling the model to capture independent category effects (e.g., a specific partner effect).

4.3.2 Z-score Standardization for Numerical Variables

For continuous variables such as *Age*, *Height*, and *YouTube Views*, we perform Z-score standardization to prevent differences in numerical scales from dominating the optimization and to ensure that the ℓ_1 regularization is applied fairly across all predictors:

$$z_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}, \tag{3}$$

where $x_{i,j}$ is the raw value of feature j for observation i, and μ_j and σ_j are the sample mean and standard deviation of feature j, respectively. After standardization, each continuous feature is centered at 0 with unit variance.

4.3.3 Missing-Value Imputation

For external numerical features with missing values, we impute using the sample mean. For missing categorical features, we assign a unified label `Unknown` and treat it as an additional category.

4.4 Abnormal Status Identification

Some contestants may withdraw mid-season due to injury or personal reasons. Such cases should not participate in the elimination calculation of that week. We introduce a status indicator $I_{i,t}$:

$$I_{i,t} = \begin{cases} 1, & \text{contestant } i \text{ withdraws in week } t, \\ 0, & \text{contestant } i \text{ competes normally.} \end{cases} \tag{4}$$

When constructing the likelihood, contestants with $I_{i,t} = 1$ are excluded from the elimination-ranking constraints in week t, while their historical observations are retained to keep the prior distribution coherent.

5 Bayesian MCMC Inference under Hierarchical Constraints

To estimate unobservable fan voting data, we develop a Bayesian MCMC inference model under hierarchical constraints. Following the latent-variable augmentation strategy[2], our model maps season-specific elimination rules into standardized likelihood functions, ensuring compatibility across all competition formats. Considering the social influence and "winner-take-all" dynamics in cultural markets[3], we adopt a Dirichlet prior to characterize the inherent sparsity of popularity distributions[4]. Due to the high-dimensionality of the posterior, we employ the Metropolis-Hastings (MCMC) algorithm [1] to robustly reconstruct the distribution of fan support within the feasible parameter space defined by the historical competition logic.

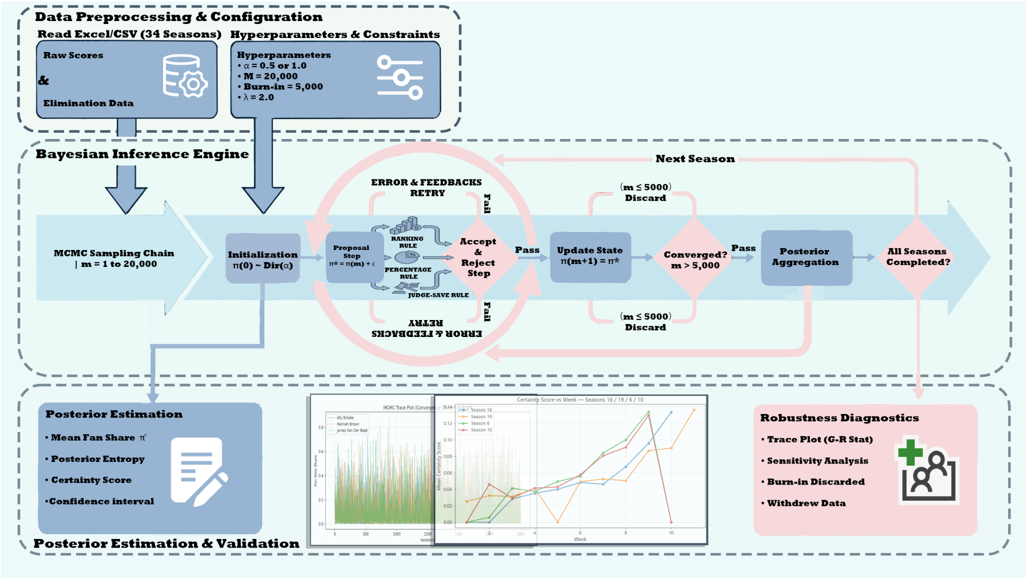

Figure 3: Overview of Model I: Bayesian MCMC inference under hierarchical constraints

5.1 Prior Distribution: Sparse Dirichlet Prior with an All-Star Adjustment

Let $\boldsymbol{\pi}_t = (\pi_{1,t}, \pi_{2,t}, \ldots, \pi_{N_t,t})$ denote the latent fan vote-share vector of all contestants in week t. Since $\boldsymbol{\pi}_t$ lies on the probability simplex, we adopt a Dirichlet prior based on conjugate-prior considerations [4]:

$$\boldsymbol{\pi}_t \sim \text{Dirichlet}(\alpha_1, \alpha_2, \ldots, \alpha_{N_t}). \tag{5}$$

Its probability density function is

$$f(\boldsymbol{\pi}_t \mid \boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{i \in \Omega_t} \pi_{i,t}^{\alpha_i - 1}, \tag{6}$$

where $B(\boldsymbol{\alpha})$ denotes the multivariate beta function.

To account for the heterogeneity of contestant pools between regular seasons and all-star seasons, we introduce an adaptive prior specification:

- **Regular seasons ($\alpha < 1$).** Because fan support typically exhibits a pronounced long-tail pattern (a small number of contestants capture a large fraction of votes), we set $\alpha_i = 0.5$. This sparse prior encourages posterior mass near simplex vertices, consistent with "winner-takes-most" dynamics.
- **All-star seasons ($\alpha = 1$).** Since contestants are previous top performers and thus more homogeneous in both skill and popularity, we adjust the prior to be uniform by setting $\alpha_i = 1$. This corresponds to a maximum-entropy stance: absent evidence, each contestant is treated as equally likely.

5.2 Likelihood Functions and Rule-Based Constraints

Because fan vote counts are unobserved, we cannot construct a standard parametric likelihood. Following the data-augmentation perspective of Albert and Chib[2], we convert the competition rules into likelihood constraints expressed via indicator functions.

Let $E_{k,t} = 1$ indicate that contestant k is eliminated in week t. We define the likelihood $L(\text{Data} \mid \boldsymbol{\pi}_t)$ to be 1 if the generated fan-support vector $\boldsymbol{\pi}_t$, combined with judge information, reproduces the observed elimination outcome; otherwise the likelihood is 0.

Across the three historical rule regimes, the combined score $C_{i,t}$ and the corresponding elimination constraints are defined as follows.

5.2.1 Case 1: Rank-based system (Seasons 1–2)

In this stage, a contestant's fate is determined by the arithmetic sum of the judge rank and the fan rank. A smaller sum implies a higher probability of advancing:

$$C_{i,t} = R_{i,t}^{\text{Judge}} + R_{i,t}^{\text{Fan}}(\boldsymbol{\pi}_t), \tag{7}$$

where $R_{i,t}^{\text{Fan}}(\boldsymbol{\pi}_t)$ is obtained by sorting $\pi_{i,t}$ in descending order. Let k denote the observed eliminated contestant in week t. The elimination constraint is

$$L_{\text{rank}}\big(\text{Data} \mid \boldsymbol{\pi}_t\big) = \mathbb{I}\left(\min_{j \in \Omega_t \setminus \{k\}} C_{j,t} \le C_{k,t}\right), \tag{8}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

5.2.2 Case 2: Percentage-based system (Seasons 3–27)

Under the percentage-based system, the combined score is the equally weighted sum of the judge score share and the fan vote share. A larger percentage score implies a higher probability of advancing:

$$C_{i,t} = J_{i,t} + \pi_{i,t}. \tag{9}$$

The elimination constraint is

$$L_{\text{pct}}\big(\text{Data} \mid \boldsymbol{\pi}_t\big) = \mathbb{I}\left(C_{k,t} < \min_{j \in \Omega_t \setminus \{k\}} C_{j,t}\right). \tag{10}$$

5.2.3 Case 3: Rank-based system with Judges' Save (Seasons 28–34)

With the introduction of *Judges' Save*, elimination proceeds in two steps: first, a bottom-two set is formed based on the combined score; second, judges decide which of the two is eliminated. To model the judges' tendency to save higher-scoring contestants, we introduce a judge-sensitivity parameter λ and specify a probabilistic likelihood.

First, the eliminated contestant k must belong to the bottom-two set. Define

$$\mathcal{B}_t = \big\{ i \in \Omega_t \,\big|\, C_{i,t} \le \min\nolimits_2 \big\{ C_{j,t} : j \in \Omega_t \big\} \big\}, \tag{11}$$

where $\min_2$ denotes the second-smallest value.

Second, if $\mathcal{B}_t = \{k, w\}$ (where w is the surviving opponent), the probability that judges eliminate k depends on the difference in judge scores $S_{k,t} - S_{w,t}$. We use a sigmoid soft constraint:

$$L_{\text{save}}(\text{Data} \mid \boldsymbol{\pi}_t) = \mathbb{I}(k \in \mathcal{B}_t) \times \frac{1}{1 + \exp\left(\lambda \left(S_{k,t} - S_{w,t}\right)\right)}. \tag{12}$$

Here λ controls how sensitive judges are to score differences: larger λ implies judges more strictly retain the higher-scoring contestant, while $\lambda \to 0$ corresponds to near-random choice. In this study we set $\lambda = 2.0$.

5.3 Posterior Inference and MCMC Sampling

By Bayes' theorem, the posterior distribution of fan support is

$$P(\boldsymbol{\pi}_t \mid \text{Data}) \propto L(\text{Data} \mid \boldsymbol{\pi}_t) \times f(\boldsymbol{\pi}_t \mid \boldsymbol{\alpha}). \tag{13}$$

Because the likelihood is a discontinuous, rule-induced truncation, the posterior does not admit a closed-form solution. We therefore apply the Metropolis–Hastings algorithm [1] for numerical simulation.

We draw $M = 20{,}000$ samples and discard the first BurnIn $= 5{,}000$ iterations as burn-in to ensure convergence to the stationary distribution. The posterior mean estimator is

$$\widehat{\pi}_{i,t} \approx \frac{1}{M - \text{BurnIn}} \sum_{m=\text{BurnIn}+1}^{M} \pi_{i,t}^{(m)}. \tag{14}$$

This approach yields not only point estimates but also uncertainty quantification via posterior standard deviations and posterior entropy computed from the retained samples.

5.4 Robustness Checks

A prerequisite for valid posterior summaries is that the Markov chain has reached its stationary distribution; otherwise, posterior means may not represent the target distribution. We therefore perform convergence diagnostics on a representative case (Season 28, Week 5) by plotting trace trajectories of $\pi_{i,t}$ over iterations (Figure 4).

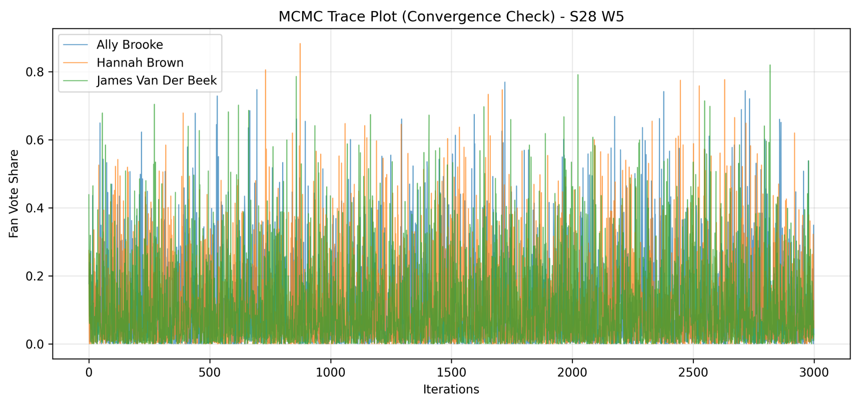

Figure 4: MCMC Trace Plot (Convergence Check) - S28 W5

As shown in Figure 4, after removing the burn-in iterations, the parameter trajectories for each contestant exhibit a stable and evenly distributed "caterpillar-like" pattern. The sampled values fluctuate rapidly around their empirical means without any systematic upward or downward drift, and the fluctuation magnitude remains broadly consistent across different segments of the chain. These features indicate good mixing of the Markov chain and suggest that the algorithm has successfully converged to the target posterior distribution; therefore, the resulting point estimates are statistically valid.

6 Results Analysis

6.1 Long-Tail Dynamics of Mean Fan Share

We next examine the evolution of the latent variable *Mean Fan Share* $\widehat{\pi}_{i,t}$ throughout a season. In a representative season, the posterior means typically exhibit a clear long-tail structure: a small subset of contestants consistently concentrate a disproportionately large share of fan support, while the majority remain in a low-support region. This reconstructed pattern aligns with the well-known social-influence dynamics in cultural markets, where popularity tends to be highly unequal and reinforcement-driven.

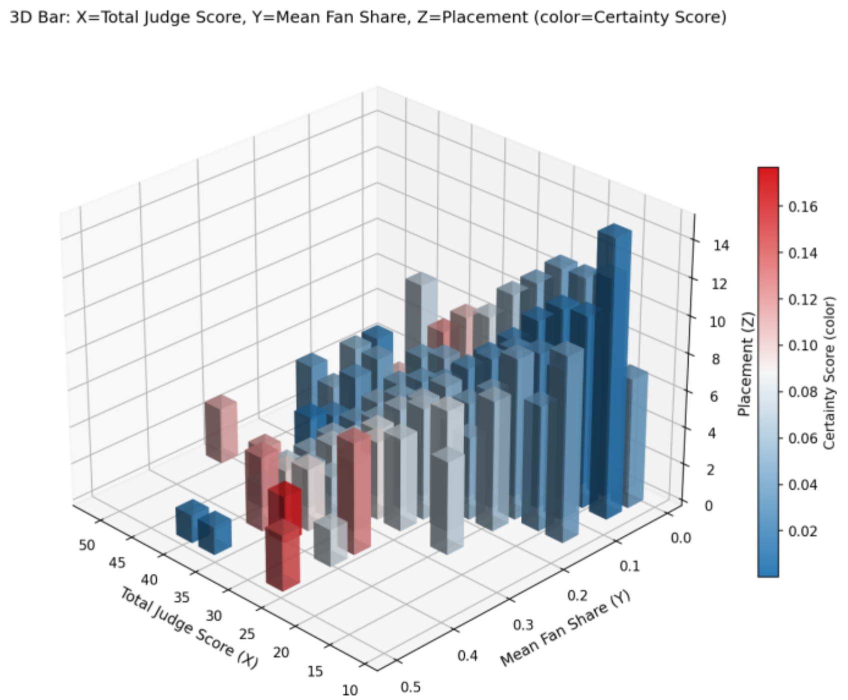

Figure 5: Spatiotemporal distribution of Mean Fan Share and the reconstructed long-tail pattern

As shown in Figure 5, a pronounced sparsity pattern can be observed. A small number of "star contestants" maintain consistently high Mean Fan Share values (e.g., $\pi_{i,t} > 0.3$), while the majority of contestants are compressed into a low-share interval of $[0.05, 0.15]$. This indicates that, under an underdetermined setting, our model successfully reconstructs the "long-tail effect" of fan voting. In other words, it effectively captures the relative popularity intensity that drives survival versus elimination, validating the rationality of introducing sparsity constraints in the prior.

Moreover, contestants with larger $\pi_{i,t}$ typically exhibit higher Certainty_Score, suggesting that the elimination logic in that week is "harder" (i.e., the constraint imposed on fan votes is stronger). For instance, when a contestant with extremely low judge scores (e.g., Bobby Bones) is not eliminated, the model is forced to assign a high and stable vote share to that contestant in order to satisfy the rule constraints, which results in a higher certainty score.

6.2 Certainty Score

In Bayesian inference, the dispersion of a posterior distribution reflects the uncertainty of parameter estimation. For contestant i in week t, although MCMC sampling provides a point estimate of fan support (the posterior mean $\pi_{i,t}$), a single value cannot characterize the confidence behind the estimate. We therefore use Shannon entropy to quantify the "disorder" of the inferred fan-vote distribution.

Let $\boldsymbol{\pi}_t = [\pi_{1,t}, \pi_{2,t}, \ldots, \pi_{N_t,t}]$ denote the posterior-mean vector of all contestants in week t, satisfying $\sum_{i=1}^{N_t} \pi_{i,t} = 1$. The posterior entropy $H(\boldsymbol{\pi}_t)$ is defined as

$$H(\boldsymbol{\pi}_t) = -\sum_{i=1}^{N_t} \pi_{i,t} \ln(\pi_{i,t}). \tag{15}$$

If the fan-vote distribution is nearly uniform, the system is in a high-disorder state and $H(\boldsymbol{\pi}_t)$ approaches its maximum, indicating that the model cannot clearly distinguish contestants and the uncertainty is high. When (and only when) one contestant has vote share 1 and all others have 0, the entropy achieves its minimum 0; when all contestants have equal vote shares, the entropy reaches its maximum $\ln(N_t)$.

6.2.1 Certainty score

Because the number of contestants N_t varies across weeks, the absolute entropy value $H(\boldsymbol{\pi}_t)$ is not directly comparable between weeks: with more contestants, the entropy upper bound increases. To remove this scale effect, we introduce a normalized Certainty Score, defined as the "deficit proportion" of the observed entropy relative to the maximum possible entropy:

$$\gamma_t = 1 - \frac{H(\boldsymbol{\pi}_t)}{H_{\max}} = 1 - \frac{-\sum_{i=1}^{N_t} \pi_{i,t} \ln(\pi_{i,t})}{\ln(N_t)}. \tag{16}$$

When $\gamma_t \to 0$, fan support is highly dispersed, the competitive landscape is extremely chaotic, and the model has the lowest confidence in the inferred structure; when $\gamma_t \to 1$, fan support is highly concentrated ("winner-takes-all"), the competitive landscape is clear, and the model exhibits very high confidence.

Although Mean Fan Share reveals popularity trends, the descriptive statistics table (Table 3) shows that the Certainty Score is generally low (mostly in the interval $[0.01,\ 0.21]$), while the Posterior Entropy remains relatively high (mean ≈ 2.119).

Table 3: Descriptive statistics of key inferred metrics

Variable	Mean	Std. Dev.	Min	25% (Q1)	Median	75% (Q3)	Max
Mean_Fan_Share	0.121	0.072	0.002	0.082	0.103	0.143	0.503
Certainty_Score	0.036	0.036	0.000	0.006	0.030	0.048	0.215
Posterior_Entropy	2.119	0.394	0.693	1.923	2.198	2.417	2.762
Metric_Value	1.683	2.708	0.002	0.092	0.135	2.995	13.141
Std_Dev	0.128	0.057	0.002	0.100	0.120	0.151	0.353
Judge_Percent	12.063	5.696	3.552	8.439	10.435	13.575	50.459
Total_Judge_Score	31.590	5.855	10.667	28.000	32.000	36.000	52.000

This numerical behavior is fully consistent with Tarantola's inverse-problem theory[6]. Because we only observe a binary elimination outcome while attempting to infer a continuous Mean Fan Share distribution, the problem is a classic underdetermined inverse problem. From Tarantola's perspective, in such information-scarce systems, the solution should not be treated as a single deterministic value, but rather as a probability distribution over the parameter space. Therefore, a low Certainty Score is not a defect of the model; instead, it honestly reflects the intrinsic uncertainty of the data: without observed vote counts, multiple Mean Fan Share configurations can explain the same elimination record.

To explore how observed outcomes reduce this uncertainty, we analyze the evolution of Certainty Score across several seasons as a function of week, as shown in Figure 6.

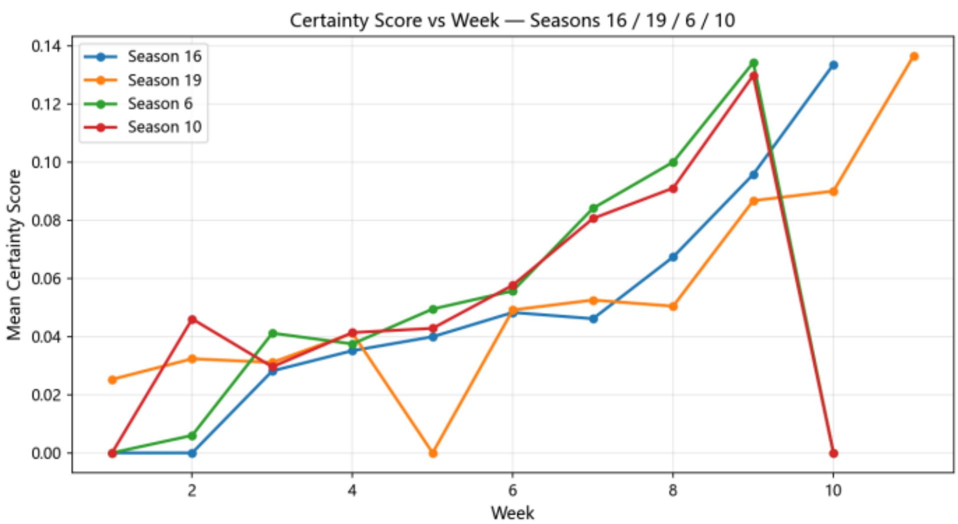

Figure 6: Evolution of Certainty Score across weeks (information gain under the maximum entropy principle)

The data indicate that the inferred certainty of Mean Fan Share tends to increase within each season, which can be explained by Jaynes' maximum entropy principle[7]:

- **High-entropy stage (Weeks 1–3).** The number of contestants is large ($N \approx 12$), so the feasible solution space has very high degrees of freedom. A single elimination event provides only a weak constraint; accordingly, the model follows the maximum entropy principle and keeps the distribution as diffuse as possible within the constrained space, leading to a large variance in Mean Fan Share and an extremely low Certainty Score.

- **Entropy-reduction stage (Weeks 9–34).** As the contestant pool shrinks, the information contained in each elimination event becomes much larger relative to the remaining solution space. The accumulation of observed outcomes yields substantial information gain, forcing the posterior distribution of Mean Fan Share to rapidly concentrate, which manifests as a marked increase in the Certainty Score.

7 Comparing Vote Aggregation Mechanisms and Counterfactual Analysis

7.1 Institutional Comparison: Rank-based vs. Percentage-based

7.1.1 Modeling objective and overall strategy

This question compares two vote aggregation mechanisms that have been used in *Dancing with the Stars*—the **rank-based rule** and the **percentage-based rule**—and examines their systematic differences in outcomes.

Because the true fan vote shares $\boldsymbol{\pi}_t$ are not publicly available and are inherently uncertain, a direct comparison based on point estimates may incorrectly attribute vote estimation error to mechanism-induced differences. To avoid this confounding, we adopt **Monte Carlo uncertainty propagation** to compare the *statistical behavior* of the two mechanisms over a plausible fan-vote space, rather than relying on a single estimated realization.

7.1.2 Assumptions

To keep the comparison interpretable, we impose the following assumptions:

H1 (Behavioral consistency). The relative preference structure of fans within the same week is stable; i.e., the fan vote distribution $\boldsymbol{\pi}_t$ does not fundamentally change when the aggregation rule switches between rank-based and percentage-based systems.

H2 (Judge independence). Judge scores reflect only professional assessment of dance quality and are independent of the fan vote aggregation rule.

These assumptions do not claim perfect independence in reality; rather, they define a comparable *counterfactual institutional environment*.

7.1.3 Mathematical description of the two scoring rules

(1) Rank-based aggregation rule. At week t, let Ω_t denote the set of active contestants. For any contestant $i \in \Omega_t$, the combined rank is

$$C_{i,t}^{\text{Rank}} = R_{i,t}^{\text{Judge}} + R_{i,t}^{\text{Fan}}, \tag{17}$$

where $R_{i,t}^{\text{Judge}}$ is the rank induced by standardized judge score $S_{i,t}$ and $R_{i,t}^{\text{Fan}}$ is the rank induced by the fan vote share $\pi_{i,t}$. The contestant with the *largest* $C_{i,t}^{\text{Rank}}$ (worst combined rank) is eliminated.

(2) Percentage-based aggregation rule. Under the percentage-based system, the combined score for contestant i is

$$C_{i,t}^{\text{Percent}} = J_{i,t} + \pi_{i,t}, \qquad J_{i,t} = \frac{S_{i,t}}{\sum_{k\in\Omega_t} S_{k,t}}. \tag{18}$$

This rule preserves **cardinal information**, so that large gaps in votes or judge scores translate directly into outcome advantages.

7.1.4 Fan-consistency metric: Spearman rank correlation

To quantify how much each mechanism aligns with fan preferences, we define a **fan-consistency index** as the Spearman rank correlation between the *combined outcome ranking* and the *pure fan ranking*:

$$\rho_t = 1 - \frac{6\sum_{i\in\Omega_t} d_{i,t}^2}{|\Omega_t|\big(|\Omega_t|^2 - 1\big)}, \tag{19}$$

where $d_{i,t}$ is the rank difference of contestant i between the combined ranking and the fan-only ranking. If, at the season level,

$$\overline{\rho}^{\text{Percent}} > \overline{\rho}^{\text{Rank}}, \tag{20}$$

then the percentage-based system is, in a statistical sense, *closer to the underlying fan preference structure*.

7.1.5 Uncertainty propagation: Dirichlet–Monte Carlo simulation

We treat $\boldsymbol{\pi}_t$ as a latent variable and model it using a Dirichlet distribution:

$$\boldsymbol{\pi}_t \sim \text{Dirichlet}(\alpha\mathbf{1}), \tag{21}$$

where α is a concentration parameter controlling the strength of uncertainty. By repeatedly sampling $\boldsymbol{\pi}_t$ for each week and computing ρ_t, we compare the average behavior of the two mechanisms over the *entire plausible voting space*, thereby reducing dependence on any single estimation path.

7.1.6 Results and mechanism-induced differences

Simulation results show that, across all seasons and weeks, the mean fan-consistency index under the percentage-based rule exceeds that under the rank-based rule by **0.2571**. This indicates that

> **the percentage-based system statistically amplifies the structural influence of fan voting.**

This conclusion can be further understood via an extreme scenario: if a contestant is ranked last by judges ($R_{i,t}^{\text{Judge}} = |\Omega_t|$) but has a fan vote share $\pi_{i,t}$ far above average, then

- Under the rank-based rule, the combined rank is almost inevitably poor;
- Under the percentage-based rule, a sufficiently large $\pi_{i,t}$ can directly offset the judge disadvantage and keep the contestant safe.

7.2 Counterfactual Analysis of Controversial Contestants

7.2.1 A feature regression model for vote dynamics

To simulate the fan-vote trajectory of contestants who *would have been eliminated* under one mechanism but *might advance* under another, we fit a weighted least squares (WLS) model:

$$\pi_{i,t} = \beta_0 + \beta_1 \pi_{i,t-1} + \beta_2 \big(S_{i,t} + \overline{S}\big) + \beta_3 \ln(t) + \beta_4 C_i. \tag{22}$$

This specification captures three key forces in reality-TV voting: **fan inertia**, **performance incentives**, and **topic-driven (publicity) effects**. The regression achieves an adjusted $R^2 = 0.483$ with a Durbin–Watson statistic of 2.109, indicating stable fit even under a high-noise environment.

7.2.2 "Comeback gap" index and representative cases

We define the comeback gap of contestant i as

$$\Delta_i = \mathrm{Rank}_i^{\mathrm{Judge}} - \mathrm{Placement}_i, \tag{23}$$

which measures how much a contestant *over-performs* in final placement relative to judge evaluation. Based on large Δ_i and high final placement, we select six high-profile contestants and run counterfactual simulations under alternative aggregation rules and with/without the Judges' Save mechanism:

Table 4: Counterfactual outcomes of high-controversy contestants under alternative mechanisms

Season	Name	Original Outcome	Alternative Aggregation	Rank + Judges' Save	Percent + Judges' Save
2	Jerry Rice	Runner-up	5th	9th	9th
3	Jerry Springer	5th	7th	7th	7th
4	Billy Ray Cyrus	5th	9th	7th	7th
11	Bristol Palin	3rd	4th	6th	4th
17	Bill Engvall	4th	8th	8th	8th
27	Bobby Bones	Champion	10th	11th	9th

The counterfactual results show that **after introducing Judges' Save, the final rankings of all controversial contestants drop substantially**. In particular, Bobby Bones (Season 27) falls from champion to the **9th–11th** range.

Significant rank suppression. The Judges' Save mechanism can effectively identify and filter out contestants with very low technical scores (judge scores) but extremely high popularity (fan votes), thereby protecting a professional baseline.

Rank-based vs. percentage-based sensitivity. Compared to the percentage-based rule, the rank-based rule penalizes low-ranked contestants more strongly. For example, Jerry Rice (Season 2) can still maintain a relatively competitive combined score under the percentage-based system due to a large fan-vote base, whereas under the rank-based system his very low judge rank largely cancels that popularity advantage, causing earlier elimination.

7.3 Recommendations for mechanism design

Synthesizing the above findings, we recommend adopting a hybrid mechanism for future seasons:

Percentage-based scoring + Bottom-two Judges' Save.

The percentage-based rule is mathematically more consistent with fan ranking, which enhances the perceived "effectiveness" of voting and encourages engagement. However, in extreme cases it may over-amplify popularity and weaken professional assessment. Introducing Judges' Save as an institutional buffer—restricted to the bottom-two contestants—provides a technical correction that prevents professional standards from collapsing. Our counterfactual simulations across multiple historical controversies confirm that this hybrid design achieves a robust balance between fan participation, show drama, and competitive integrity.

8 Identifying the characteristics of celebrities' success and failure

8.1 A Lasso-based multi-factor influence attribution model

After obtaining posterior estimates of the latent variable "fan support" via Bayesian inference, we next investigate which observable features drive (i) judges' evaluations and (ii) audience voting. During data preprocessing, we introduced a large collection of high-dimensional sparse covariates (e.g., 60 professional partners, 26 occupations, and 45 home states). Consequently, the ratio between feature dimension p and sample size N increases substantially. Under such settings, conventional ordinary least squares (OLS) is prone to multicollinearity and overfitting.

Following Tibshirani (1996), the Least Absolute Shrinkage and Selection Operator (Lasso) introduces an ℓ_1-regularization penalty that shrinks the coefficients of non-informative variables toward zero, thereby performing automatic feature selection. This property aligns with our objective of identifying core drivers from a large set of potentially noisy predictors.

We build two separate regression models to explain the judges' score share $J_{i,t}$ and the fan-support index $F_{i,t}$, respectively. Let $y_{i,t}$ denote the response variable, and let $\mathbf{x}_{i,t} \in \mathbb{R}^p$ be the standardized feature vector for contestant i in week t, including numerical attributes (e.g., age, YouTube popularity, and height) as well as one-hot encoded categorical variables.

The Lasso estimator solves

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i,t} \left(y_{i,t} - \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta} \right)^2 + \lambda \, \|\boldsymbol{\beta}\|_1 \right\}, \tag{24}$$

where $\lambda \geq 0$ controls the strength of regularization and $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|$ is the ℓ_1 norm.

$$\boldsymbol{\beta} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2N} \sum_{i,t} \left(y_{i,t} - \beta_0 - \sum_{j=1}^{p} x_{i,t,j}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}. \tag{25}$$

Here $\lambda \geq 0$ is a tuning parameter that controls the sparsity level of the model. Following the statistical learning perspective in Hastie et al. (2009), the feasible region induced by the ℓ_1 penalty is, geometrically, a diamond-shaped polytope centered at the origin. When the residual sum-of-squares contours become tangent to this constraint set, the tangency point tends to occur on coordinate axes, which forces some coefficients β_j to be exactly zero[9], thereby removing noisy or non-informative variables.

8.2 Results Analysis

We use five-fold cross-validation (5-fold CV) to select the optimal regularization parameter λ, and conduct a comparative analysis between the *judge model* and the *fan model*.

Table 5: Model selection and feature sparsity for the judge model and fan model

Model type	Dependent variable	Best λ	R^2 (test set)	Feature retention rate
Judge model	$J_{i,t}$	0.0023	0.2751	19% (123/152 retained)
Fan model	$F_{i,t}$	0.0002	0.0417	76% (36/152 retained)

8.2.1 Dimension Disparity between Judges and Fans

The normalized coefficient comparison bar chart visualizes the magnitudes of standardized coefficients across covariates, thereby avoiding scale inconsistencies between judge scores and fan-vote shares.

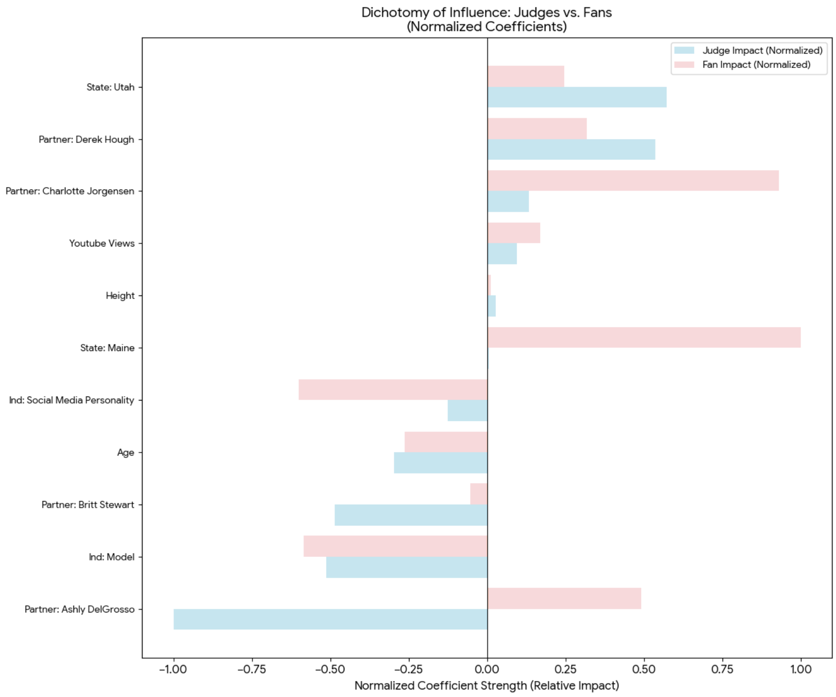

Figure 7: Normalized coefficient comparison between the judge model and the fan model

The *Partner* variable (Partner: Derek Hough) receives an exceptionally large positive weight in the judge model. This finding is consistent with O'Connor & Cheema (2018), who argue that judge scoring may exhibit a *halo effect*, whereby trust in elite professional dancers is implicitly transferred to their celebrity partners[10].

The variable *Age* has a markedly negative coefficient in the judge model (-2.19), suggesting that judges penalize older contestants for perceived physical limitations;

by contrast, *Age* is nearly neutral in the fan model (-0.01). This pattern aligns with the reality-TV psychology evidence reported by Reiss & Wiltz (2004): audiences often overlook technical flaws due to an *underdog expectation* and thus do not treat age as a salient barrier to voting[14].

8.2.2 The Social Media Paradox

As shown in Figure 8, we test the common hypothesis that "traffic equals legitimacy" by comparing the standardized coefficients of *Social Media Personality* and *YouTube Views*. The contrast reveals a deeper logic of fan voting that cannot be explained by exposure volume alone.

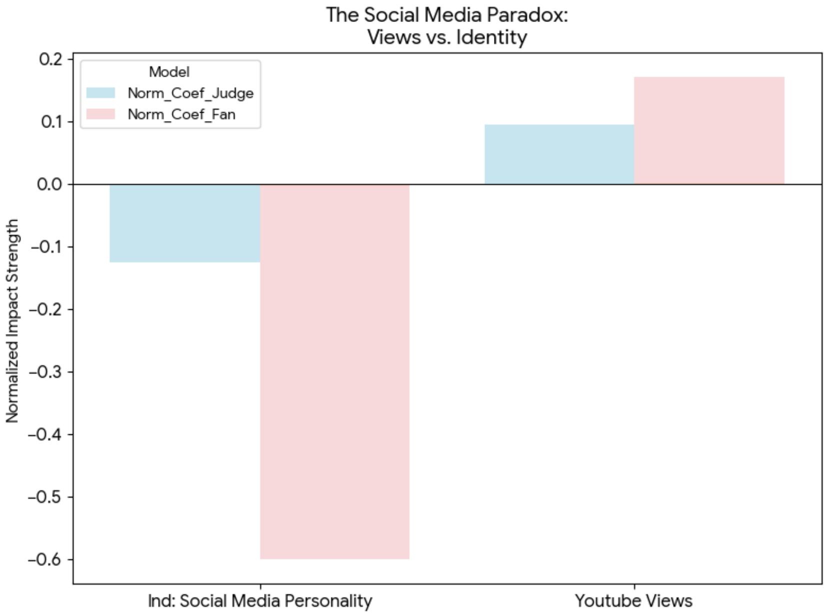

Figure 8: The Social Media Paradox: Views vs. Identity

Lasso regression reveals contrasting effects of digital presence on fan voting: *YouTube Views* exert a positive influence $+0.007$, whereas *Social Media Personality* shows a negative impact -0.026. This disparity can be attributed to "slacktivism" [11], where low-cost online engagement fails to translate into high-cost registration-based voting, and potential in-group bias among the shows older television audience [12]. Overall, the analysis characterizes the shows dual logic: the judging system embodies technical rationality and elitism, prioritizing professional metrics and credibility; conversely, the fan system is affect-driven and stochastic, valuing regional identity while reacting negatively to pure "influencer" labels. These findings motivate the design of the Logistic Dynamic Weighted System (LDWS) to better balance these competing forces.

9 A New System: Logistic Dynamic Weighted System

The above analysis indicates that rank-based or percentage-based rules cannot simultaneously balance **fan engagement incentives** and **professional fairness** across different stages of a season. If judges dominate too early, high-visibility contestants may be eliminated prematurely, reducing viewers' willingness to vote and weakening audience retention. Conversely, if fan influence remains dominant late in the season, the outcome may be driven by a *popularity landslide*, undermining the professional credibility of the competition. Although the "having judges choose which of the bottom two couples" rule can partially protect technically strong contestants, it only acts *after* a bottom group is formed and may be too late to correct distortions created earlier.

To resolve this tension, we propose the **Logistic Dynamic Weighted System (LDWS)**. Its design follows three principles:

- **Stage adaptivity**: the relative influence of fans and judges should change as the season progresses;
- **Diminishing marginal influence**: the marginal impact of fan votes should decrease as vote totals grow, preventing extreme concentration;
- **Interpretability & implementability**: the mechanism should remain transparent and easy to implement using existing data.

9.1 Overall structure

Let $S_{i,t}^{\text{total}}$ denote the composite score of contestant i in week t. LDWS is defined as

$$S_{i,t}^{\text{total}} = w(t)\, J_{i,t} + \big(1 - w(t)\big)\, \pi_{i,t}, \tag{26}$$

where $\tilde{J}_{i,t}$ is the normalized judge-score share, $\pi_{i,t}$ is the transformed-and-normalized fan-vote share, and $w(t) \in (0,1)$ is a week-dependent judge weight. This preserves the intuitive weighted-sum form while enabling institutional-level dynamic adjustment through the redesigned $w(t)$ and the fan-vote term.

9.2 Nonlinear transformation of fan votes

To suppress the "Matthew effect" in which a superstar's popularity overwhelms the ranking, we apply a concave transformation to raw vote counts $V_{i,t}$:

$$\Phi\big(V_{i,t}\big) = V_{i,t}^{\gamma}, \qquad 0 < \gamma \le 1. \tag{27}$$

We then normalize within each week:

$$\pi_{i,t} = \frac{\Phi\big(V_{i,t}\big)}{\sum_{k=1}^{n_t} \Phi\big(V_{k,t}\big)}, \tag{28}$$

where n_t is the number of active contestants in week t. In our baseline setting we take $\gamma = 0.5$ (square-root transformation). Intuitively, fan voting remains positively rewarded, but its *marginal contribution* decreases with scale, limiting distortions created by extreme vote concentration.

9.3 Dynamic judge weight via a logistic function

To achieve a smooth transition across stages, we model the judge weight as a logistic curve:

$$w(t) = w_{\min} + \frac{w_{\max} - w_{\min}}{1 + \exp\big(-k(t - t_0)\big)}. \tag{29}$$

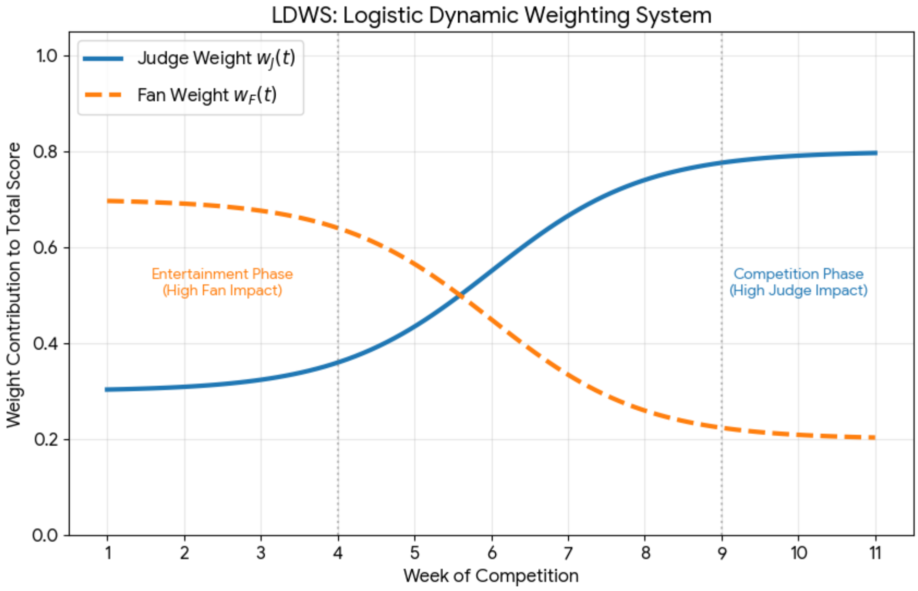

Figure 9: LDWS: Logistic Dynamic Weighting System weights over weeks

Parameters are defined as follows:

- $w_{\min}$: early-season judge weight ,we use 0.3;
- $w_{\max}$: upper bound of late-season judge weight ,we use 0.8;
- t_0: midpoint week of the transition;
- k: steepness parameter controlling how abruptly the weight changes.

Thus, in early weeks ($t \ll t_0$), fan votes dominate, while the judge weight increases steadily and becomes the main determinant toward the finale.

9.4 Counterfactual Validation on Controversial Contestants

To examine LDWS performance in suppressing extreme fan effects and improving overall fairness, we select several historically controversial contestants. We then compare their real placements under the original voting rules with their counterfactual placements predicted under LDWS.

Table 6: Counterfactual validation on controversial contestants: real placement vs. LDWS model placement

Season	Name	Real Placement	Model Placement
2	Jerry Rice	2	4
3	Jerry Springer	5	6
4	Billy Ray Cyrus	5	11
11	Bristol Palin	3	3
15	Bristol Palin	9	7
17	Bill Engvall	4	10
27	Bobby Bones	1	6

The LDWS model systematically mitigates the disproportionate advantage of contestants who rely solely on fan bases by applying "stage-dependent professional weighting" and nonlinearly compressing the marginal influence of fan votes. For instance, Bobby Bones (S27), the original champion, drops to 6th place under LDWS, whereas Bristol Palin—who maintained stable technical scores—retains her high ranking. This indicates that LDWS precisely identifies and suppresses abnormal advancement paths driven purely by popularity, rather than indiscriminately penalizing high-visibility participants. From a production standpoint, this mechanism allows the show to retain high-profile contestants early to maximize viewership while ensuring that late-stage outcomes remain technically defensible, thereby enhancing long-term credibility and competitive fairness.

9.5 Interpretation and practical implications

LDWS improves the reality-competition mechanism in a structured way:

- **Boosts fan engagement**: higher early-stage fan weight makes voting effects perceptible, strengthening participation and retention.
- **Protects late-stage credibility**: gradually increasing judge weight ensures that finalists meet a professional performance baseline.
- **Reduces popularity-induced distortion**: the nonlinear vote transform weakens the dominance of explosive popularity.
- **Transparent and implementable**: the system uses only existing judge scores and fan votes, with parameters that have clear operational meanings.

10 Sensitivity Analysis

To evaluate the potential subjectivity introduced by prior selection, we conduct a sensitivity analysis on the hyperparameter α. We compare the original model which incorporates a "winner-takes-most" sparse prior $\alpha = 0.5$ for regular seasons and $\alpha = 1.0$ for All-Star seasons against a control model using a uniform prior ($\alpha' = 1.0$) for all instances. Despite the shift in prior assumptions, the posterior mean fan-support vectors exhibit an extremely high Spearman rank correlation ($\rho = 0.972$). As illustrated in Figure 10, the scatter points concentrate tightly around the $y = x$ diagonal, indicating that the inferred contestant rankings remain materially unchanged. This high degree of agreement demonstrates that the elimination constraints within the likelihood $L(\text{Data} \mid \boldsymbol{\pi}_t)$ provide sufficient information to override prior specifications, confirming the robustness of our Bayesian framework.

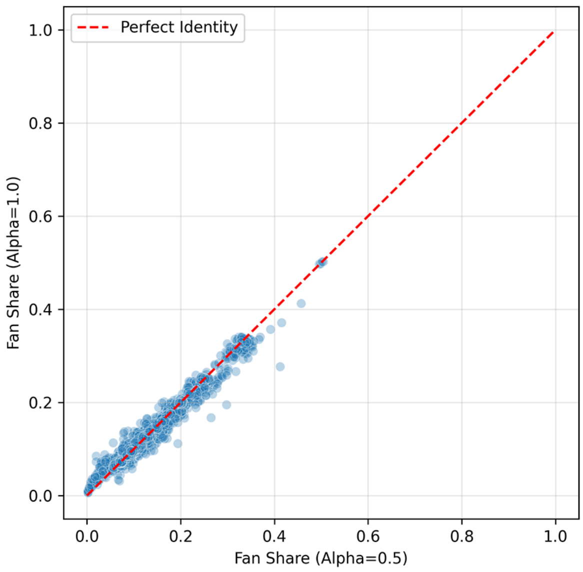

Figure 10: Prior Sensitivity Analysis (Prior Robustness)

11 Conclusion

11.1 Model strengths and limitations

Under the constraint that true fan-voting records are unobservable, this paper reconstructs the distribution of contestants' Mean Fan Share using only elimination outcomes and judge scores. Across all seasons and weeks, the model accurately reproduces historical eliminations. Compared with ad-hoc empirical rules or direct regression-based prediction, our approach provides clearer uncertainty quantification and interpretability, while naturally adapting to different competition formats.

Nevertheless, to maintain tractability we assume fan voting is independent across weeks, and thus do not explicitly model the time dependence of popularity or the inertia of public opinion. In addition, judge behavior is represented in a simplified parametric form. These assumptions may understate the true complexity of the underlying behavioral system.

11.2 Future research directions

Future work can be extended along two main directions. First, fan-behavior modeling can incorporate time-series or state-space structures to capture the dynamic evolution of support. Second, at the mechanism-design level, game-theoretic or multi-agent models can be introduced to describe interactions between judges and fans, enabling a more systematic assessment of the long-run impacts of different rules.

Overall, this study demonstrates how, in information-constrained real-world settings, mathematical modeling and statistical inference can be used to extract structural regularities of institutional operation, providing a quantifiable and generalizable analytic framework for improving reality-show competitions and other public-voting-based contest mechanisms.

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Who Really Keeps Dancing?

Dear Producers,

To resolve the historical tension between technical judging and fan popularity, our team has completed a rigorous quantitative study of all 34 seasons of Dancing with the Stars. We have reconstructed the hidden dynamics of the show's scoring system using advanced statistical modeling to provide actionable recommendations for future seasons.

Summary of Our Modeling Process

Inverse Estimation of Fan Votes:
Because fan votes are unobservable, we developed a Bayesian MCMC inference model with hierarchical rule constraints to reconstruct latent _Mean Fan Share_ from elimination outcomes and judge scores，the model achieved 100% consistency with historical eliminations.

Counterfactual Mechanism Analysis:
Based on the inferred fan support, we simulated alternative scoring systems and showed that the percentage-based rule amplifies fan influence by an average of 25.7% compared to rank-based systems, providing a quantitative explanation for controversial outcomes such as Bobby Bones.

Feature Attribution:A Lasso regression identified key drivers of success. We quantified a strong “halo effect” from elite pro-partners onjudge scores and revealed a social media paradox: while YouTube exposure has a small positive effect, influencer-type contestants exhibit lower effective vote conversion.

Proposed Solution (LDWS):
We proposed the Logistic Dynamic Weighted System (LDWS), which dynamically adjusts fan–judge weights via a logistic function—emphasizing fan influence in early stages to boost engagement and shifting toward judge dominance in the finale to ensure technical fairness.

Recommendations for Future Seasons

Nonlinear Vote Transformation: We recommend a square-root transformation of raw fan votes to suppress the "Matthew Effect," ensuring that a single dominant fanbase does not make the competition outcome a foregone conclusion.

Adopt LDWS for Strategic Balance: Implement the LDWS to retain popular stars early for ratings while protecting the show's professional credibility at the finish line.

Let’s give the fans the fairness they deserve and the stars the spotlight they’ve earned. Together, we can make the next season of _Dancing with the Stars_ the most thrilling and talked-about one yet!

Sincerely,
Team 2614173

Use of AI

AI tool used: Gemini3 pro
Date of use: January 29 - February 2, 2026

1. **Data cleaning and feature extraction guidance.**
   • **Input:** Field-structure description of the raw dataset `2026_MCM_Problem_C_Data.csv`.
   • **Output:** Python snippets to extract judges' scores by *Season* and *Week*.
   • **Adopted:** NaN imputation logic and a wide-to-long restructuring approach.

2. **Multi-format data integration script.**
   • **Input:** How to merge CSV files with an exported Excel file.
   • **Output:** Examples using `pandas.merge` and `pandas.concat`.
   • **Adopted:** A left-join scheme keyed by `celebrity_name` and `season`.

3. **LaTeX math syntax proofreading.**
   • **Input:** Draft equations for complex statistical models.
   • **Output:** Academic-standard LaTeX formatting (e.g., `\sum`, $\bar{x}$, `\frac`).
   • **Adopted:** Standardized math syntax throughout the paper.

4. **Paper structure and organization suggestions.**
   • **Input:** Request for a standard MCM paper structure and logical flow.
   • **Output:** A recommended end-to-end framework from the Summary Sheet to Strengths & Weaknesses.
   • **Adopted:** The layout of *Assumptions and Justifications*.

5. **Data visualization style guidance.**
   • **Input:** Suggestions for plots that illustrate "contestant advancement trends."
   • **Output:** Recommendations such as a Sankey diagram or a stacked area chart, with Matplotlib style parameters.
   • **Adopted:** A colorblind-friendly palette.

6. **Automation and runtime optimization.**
   • **Input:** A slow Python loop implementation.
   • **Output:** Optimization advice replacing explicit `for` loops with NumPy vectorization.
   • **Adopted:** Vectorized improvements in large-scale simulation code.

7. **Academic wording refinement.**
   • **Input:** Colloquial draft sentences (e.g., "The judges like some dancers more").
   • **Output:** More formal phrasing (e.g., "Numerical evidence suggests a potential bias in judges' scoring distributions...").
   • **Adopted:** Wording improvements in approximately 30% of key argument paragraphs.

8. **Sensitivity analysis code framework.**
   • **Input:** A generic one-parameter sensitivity analysis Python template.
   • **Output:** A code skeleton with perturbation settings and plotting functions.
   • **Adopted:** A modified version was implemented to test model stability under parameter perturbations.

9. **Summary Sheet logic refinement.**
   • **Input:** Multiple conclusions in the abstract draft.
   • **Output:** Suggestions to reorganize the abstract using the "Problem–Method–Result–Conclusion" logic.
   • **Adopted:** Sentence structures that improved the clarity and impact of the Summary Sheet.

10. **Reference formatting conversion.**
    • **Input:** Non-standard web links and journal names.
    • **Output:** Standardized BibTeX or APA-style references.
    • **Adopted:** Used to organize the References section.